OVAI — Methodology, Downstream Analysis, and Competition Presentation Plan

Prepared from a review of the OVAI-main codebase (100 scripts, reports/, configs/, production_artifacts/) and the project outline supplied by the team. Companion document: OVAI_Errors_Suggestions_and_Notes.md, which contains the detailed QA findings referenced throughout this plan.

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1. The Problem, Stated Precisely

Ovarian cancer is the deadliest gynecologic malignancy in the United States. The American Cancer Society projects roughly 21,000 new diagnoses and 12,450 deaths in the U.S. in 2026, against a lifetime risk of about 1 in 91. The reason it is so lethal is almost entirely a detection-timing problem, not a treatment problem: five-year survival is roughly 92% when the disease is caught while still confined to the ovary, but the large majority of cases are still found after the cancer has already spread, where survival falls to the 30–40% range. This is the single fact that justifies the entire project — a screening or triage tool only matters if it moves diagnosis earlier in the disease course. It is important to be precise about *why* earlier detection hasn't already been solved, because that is where the real "friction" the competition is asking about lives:

- Single-analyte CA-125 is not sensitive or specific enough on its own, especially in premenopausal women and in non-serous histologies (mucinous, clear-cell, endometrioid), and it rises with many benign conditions (endometriosis, fibroids, pregnancy).
- Large randomized screening trials have already failed to show a mortality benefit from annual CA-125 and transvaginal ultrasound in the general population (UKCTOCS and PLCO). This is why no medical body currently recommends population-wide ovarian cancer screening for average-risk women.
- The three FDA-cleared multi-marker blood tests that exist today — OVA1 (2009), ROMA (2011), and Overa (2016) — were deliberately *not* cleared as screening tests. They carry an FDA-mandated black-box warning restricting their intended use to risk-stratifying a mass a physician has *already* found on imaging, before referral to a gynecologic oncologist. Marketing any of them for population screening is off-label.
- The workflow gap is not "no test exists," it is "no test exists at the point in care where it would change a decision that isn't already being made." A test that duplicates what CA-125 + ultrasound already does at the adnexal-mass-workup stage adds little. A test that could plausibly change practice would need to either (a) beat the existing adnexal-mass triage tests

head-to-head, or (b) create a *new* decision point earlier in the pathway — e.g., in symptomatic but not-yet-imaged women in primary care — that doesn't exist today.

This reframing matters for how OVAI should be positioned. The prompt asks for "genuine understanding of the scope of the problem," and the strongest, most defensible answer is: *the bottleneck is not a missing algorithm, it is a missing low-friction entry point into an already-crowded, well-regulated diagnostic pathway.* Sections 7–8 below lay out a concrete plan for that entry point; Section 4 shows exactly what the current model can and cannot yet claim.

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2. Recommended Clinical Positioning

Given the above, the team should explicitly choose (and say out loud, to judges) which claim OVAI is making. There are three tiers, in increasing order of ambition, evidence requirement, and regulatory difficulty:

Tier	Claim	Comparable precedent	Evidence needed
1. Adnexal-mass triage adjunct	"For a mass already found on imaging, this molecular panel helps decide low-risk vs. refer-to-gyn-onc"	OVA1 / ROMA / Overa (all cleared on exactly this claim)	Head-to-head or non-inferiority data vs. CA-125/HE4-based comparators, in the adnexal-mass population
2. Symptomatic-population triage	"For a woman with persistent GI/pelvic symptoms in primary care, before any imaging has been ordered,	No FDA-cleared precedent exists here — genuine white space, but also genuinely harder to validate	Prospective symptomatic-cohort study, calibrated against a validated symptom index

	this flags who should be worked up sooner"		
3. Average-risk population screening	"This screens asymptomatic women the way mammography screens for breast cancer"	UKCTOCS/PLCO already tested this idea with CA-125 + ultrasound and found no mortality benefit	Would require a large prospective mortality-endpoint trial; not realistic on the project's current timeline or evidence base

Recommendation: lead with Tier 1 now, and frame Tier 2 as the funded next step. Tier 1 is the claim the current GSE94533 cohort actually supports (it is a mix of women with a diagnosed adnexal mass undergoing workup), it has direct FDA and reimbursement precedent to point to, and it gives judges/investors a believable near-term commercial path (compete with or complement OVA1/ROMA/Overa) rather than an open-ended "we will screen the whole population" claim that the field's own largest trials have already shown doesn't work as a mortality-reduction strategy. Tier 3 should not appear in competition materials as a near-term claim.

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3. What Data OVAI Actually Has

Source	Role	N (current build)	Modality	Patient-linked to GSE94533?
GSE94533 (GEO)	Canonical training/evaluation cohort	179 (96 cancer, 83 non-cancer: 46 benign, 22 borderline, 15 control)	Serum miRNA-seq (2,582 miRNAs) + GEO metadata	— (this *is* the base cohort)

eLife Supplementary Dataset 2	Historical reconstruction of the same patients	179	Batch-adjusted log10 miRNA (192-miRNA subset)	Same patients, not an independent cohort
eLife PMML / Table 2	Original published 14→7→2 tanh neural-network model + its predictions	179 (reference only)	Model artifact, no new patient rows	Same patients
GSE106817 / GSE113486 / GSE113740 (GEO)	External serum-miRNA stress tests	3,108 / 972 / 1,817	3D-Gene GPL21263 miRNA array	Independent cohorts; no verified patient overlap with GSE94533
MethylBERT / HRA011451 supplement	Independent methylation benchmark	1,872 (cancer vs. healthy); 1,323 (early-stage vs. healthy)	Blood/cfDNA methylation, genomic-coordinate features	Independent cohort; not patient-linked to GSE94533
TCGA-OV	Tumor-tissue reference cohort	299 (RNA/miRNA/clinical)	Tissue RNA-seq, miRNA-seq, CNV, clinical (patient-barcode linked within TCGA)	Not linked to GSE94533; tissue, not serum

dbGaP phs000897	Controlled-access early-stage serum cohort	Not yet obtained	Unknown pending authorization	Would be a genuinely independent early-stage validation cohort if access is secured

The project has already committed to a rule worth keeping and stating explicitly to judges: never concatenate rows from different studies to fabricate a "multimodal" patient. This is good scientific hygiene and is exactly the kind of discipline a healthcare-focused judging panel should be shown, not just told about — see Section 9.

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4. Current Model Results — Reported Honestly, Side by Side

This is the most important section for how the project should talk about its own results. Several different pipelines in the repository target the same underlying question (does the frozen Top-5-miRNA + Age + CA-125 signature separate early cancer from non-cancer?) but on different patient subsets, and they produce meaningfully different headline numbers. Judges and technical reviewers will ask "compared to what population?" — the team should already have the answer ready.

Endpoint definition	N	Model	ROC-AUC	Balanced accuracy	Sensitivity	Specificity
Full cohort: all 96 invasive cancers vs. all 83 non-cancer (benign+borderline+control)	179	14 miRNA + Age + CA-125, logistic	0.813	74.0%	—	—
Full cohort, calibrated SVM, inner-fold-tuned threshold	179	Calibrated SVM	0.808	77.1% (acc. 76.5%)	—	—

Narrower endpoint: Stage I/II cancer only vs. benign/control only (borderline and Stage III/IV excluded)	113 (52 pos / 61 neg)	Top-5 fixed miRNA signature + Age + CA-125, logistic (no interactions)	0.899	86.6%	84.6%	88.5%
Same 113-patient endpoint, with 2 interaction terms (this is the model shipped in production_artifacts/ and the Streamlit portal)	113	Engineered logistic (production model)	0.900	86.4%	82.7%	90.2%

Read this table carefully before it goes in a pitch deck. The 0.90 AUC / 86%/83%/90% headline number is real and is a legitimate nested cross-validated estimate — but it is measured on an *easier* 113-patient subset that excludes both borderline tumors (the diagnostically hardest category) and Stage III/IV cancers. The 179-patient number (0.81 AUC) is the one that answers "does this work across the whole clinical spectrum this cohort represents." Present both, labeled clearly, rather than leading only with the larger number — see OVAI_Errors_Suggestions_and_Notes.md §1 for why the gap between these two numbers is bigger than typical CV noise and what it implies about the feature-selection process, and §1 also flags that the two interaction terms contribute essentially nothing to AUC (+0.0003) while trading sensitivity for specificity versus the simpler model — worth knowing before deciding which model to actually put into the demo.

External validation — the current bottleneck, stated plainly

External cohort	N	Ranking ROC-AUC	Sensitivity at frozen threshold	Specificity at frozen threshold
GSE106817	3,108	0.82	0%	100%
GSE113486	972	0.83	0%	100%
GSE113740	1,817	0.77	0%	100%

The model trained on GSE94533, applied unchanged to three independent serum-miRNA cohorts, ranks cases reasonably well (AUC 0.77–0.83) but classifies literally every external sample as non-cancer at the threshold learned from the training cohort. This is a textbook cross-platform/cross-lab measurement-scale shift, not a failure of the underlying biology — the

ranking signal survives, the absolute scale does not. This is the single most important open scientific problem in the project, and it should be presented to judges as exactly that: a named, understood, and fundable problem, not something to omit. A diligence-savvy panel will be more impressed by a team that found and explained this than by one that didn't look.

Independent-modality benchmarks (not yet fused with the serum model)

- Methylation (HRA011451/MethylBERT supplement, blood/cfDNA): cancer vs. healthy — 88.1% accuracy, 86.9% balanced accuracy, 0.941 AUC (N=1,872). Early-stage vs. healthy — 84.1% accuracy, 77.3% balanced accuracy, 0.861 AUC (N=1,323). Strong signal, but this is a separate cohort and not patient-linked to GSE94533 — it demonstrates that circulating methylation carries independent diagnostic information, not that it improves the miRNA model.
- TCGA-OV tumor stage (Stage I/II vs. III/IV, N=299, 22 vs. 277): 78.9% accuracy is *below* the 92.6% majority-class baseline (balanced accuracy 53%, AUC 0.49). This is not a cancer-detection result and should never be cited as one — it is a separate, currently-unsuccessful research question (tumor staging from molecular data) that happens to share a data source with the main project.
- The gated multimodal fusion network (src/models/gated_fusion.py) is an engineering scaffold, exercised only against random synthetic tensors in a unit test (scripts/smoke_test.py). It has never seen a real patient. It should be described in competition materials as "architecture built and unit-tested, awaiting a patient-linked multimodal cohort to train on" — not as a working multimodal model.

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5. Rigor Already In Place (worth highlighting to judges)

Not everything here is a gap. Several choices reflect real methodological maturity and are worth stating explicitly rather than assuming judges will notice them in the code:

- Canonical evaluators use nested cross-validation (outer 5-fold for the reported metric, inner 3-fold for hyperparameter selection) with fixed seeds, rather than a single train/test split.
- A hard rule against fabricating multimodal patient rows across unrelated studies is written into the README and enforced by validation checks in the data-loading code.
- CPTAC proteomics was investigated and explicitly declined as a fusion source because no verified patient-level, blood-specimen link to GSE94533 exists — a disciplined "no" that

many teams under deadline pressure would not have made (see reports/cptac_acquisition_audit.md).

- 58 candidate external GEO cohorts were systematically screened against platform, specimen, and phenotype criteria, with the screening logic and rejection reasons saved (reports/potential_external_datasets.md, reports/remaining_queue_screen.md).
- Borderline tumors are kept as their own category rather than being folded into either "cancer" or "non-cancer" for convenience — clinically correct, since borderline ovarian tumors are a genuinely distinct, non-invasive entity.

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6. Downstream Analysis Plan — Phased Roadmap

Phase 0 — Before the submission deadline (days)

1. Pick one canonical headline result and present the full endpoint table from Section 4 next to it, not just the best number.
2. Add a calibration plot (predicted probability vs. observed frequency) for the production model — currently no calibration check exists anywhere in the repo.
3. Run a decision curve analysis (net benefit vs. threshold probability) for the production model against "treat/refer all" and "refer none" baselines. This is the standard way clinicians and clinical judges evaluate whether a risk score is actually useful at plausible decision thresholds, and it is a much stronger artifact for a healthcare-focused competition than ROC-AUC alone.
4. Re-run the frozen 113-patient evaluation with the feature-selection step performed inside each outer fold (see OVAI_Errors_Suggestions_and_Notes.md §1) and report that number alongside the current one, so the team can honestly say what the signature's performance looks like under a stricter protocol.

Phase 1 — 0–3 months: rigor hardening

- Move the remaining global (pre-split) feature-availability filters inside the training fold in the canonical GSE94533 and methylation evaluators.
- Add an explicit sample/patient-ID overlap audit between GSE94533 and every external cohort (GSE106817/113486/113740) — currently assumed, not verified.
- Formalize the subgroup analysis already started in reports/error_analysis.md (by stage, histology, missingness) into a standing report that is regenerated with every model change,

and add histology-stratified sensitivity as a first-class reported metric (see Section 4's note on clear-cell/endometrioid histology below).
- Build the single canonical run manifest recommended in the team's own reports/current_pipeline_audit.md (one script, one command, one seed, one output file per reported number).

Phase 2 — 3–6 months: closing the external-validation gap

- Apply for dbGaP phs000897 access now — this is the only currently-identified early-stage serum cohort that could serve as genuine independent validation, and controlled-access approvals commonly take 4–8+ weeks, so the application timeline should start well before it is needed.
- Source or prospectively collect an HE4-linked cohort. HE4 is a component of both ROMA and Overa; without it, OVAI cannot run a head-to-head comparison against the two most relevant FDA-cleared competitors, which will be one of the first questions a knowledgeable judge or investor asks.
- Investigate platform harmonization (e.g., ComBat, removeBatchEffect) only once a serum cohort with real batch/run metadata is available — the audit already correctly notes that applying such corrections without batch labels amounts to fitting noise. If reaching out to the GSE94533 authors for original run-batch metadata is possible, that is the most useful single request the team could make.

Phase 3 — 6–12 months: real multimodal fusion and longitudinal modeling

- Only build patient-linked multimodal training rows once a cohort exists where the same patient has both serum miRNA and a second modality (methylation, HE4, imaging) drawn at the same visit. Until then, keep methylation and TCGA as independent-cohort benchmarks, exactly as the project already does.
- Consider a longitudinal/trend model in the style of the Risk of Ovarian Cancer Algorithm (ROCA) used in UKCTOCS — modeling the *rate of change* of CA-125 (and potentially the miRNA signature) over repeated draws in the same patient, rather than a single time point. This is a well-precedented idea in this exact disease area and is a natural extension once a cohort with repeat draws exists.

Phase 4 — 12+ months: prospective clinical validation

- Design a prospective single-platform cohort study in the Tier-1 population (women with an adnexal mass headed to workup), matching the evidence design that supported OVA1/ROMA/Overa, so the resulting data can support a comparable regulatory claim.

7. Regulatory and Reimbursement Pathway (the "friction" section)

This is where the competition's stated focus — "the friction of getting innovation into the healthcare system" — is most directly addressed, and it is the section most pitch decks in this space skip entirely.

Two realistic pathways exist, and they are not mutually exclusive over time:

1. Laboratory-Developed Test (LDT) via a single CLIA-certified reference laboratory. As of the FDA's formal rule vacatur on September 19, 2025 (following the March 2025 federal court ruling that FDA lacks statutory authority to regulate LDTs as devices), LDTs are regulated under CLIA (administered by CMS), not the FDA device framework. This is currently the lower-friction, faster path to a real-world clinical offering: OVAI could be validated and offered as a send-out LDT from one reference lab without a 510(k)/PMA/De Novo submission. The tradeoff is narrower distribution (one lab performs the test) and no FDA clearance to point to with payers.
2. FDA in vitro diagnostic clearance (510(k) or De Novo). This is the path OVA1 (2009), ROMA (2011), and Overa (2016) all took, and it is the path to broad multi-lab distribution and the strongest payer credibility. For an AI/ML-based test specifically, the FDA finalized guidance on Predetermined Change Control Plans (PCCPs) in August 2025: a PCCP lets a manufacturer pre-specify how the model will be updated after clearance (e.g., periodic retraining on new data) so that in-scope updates don't require a brand-new submission each time. This directly reduces the "friction" of maintaining an AI-based diagnostic over time and should be built into the regulatory strategy from the start rather than retrofitted later.

Reimbursement note: existing payer medical policies for OVA1/ROMA/Overa explicitly restrict coverage to adnexal-mass workup and explicitly exclude screening asymptomatic women or monitoring treatment — this is direct, current evidence for why Tier 1 positioning (Section 2) is the commercially realistic starting claim, and why claiming a screening indication before the evidence exists would likely be denied coverage even if a lab offered it as an LDT.

Recommended near-term regulatory move: an early, informal engagement with a regulatory consultant or an FDA pre-submission ("Q-Sub") meeting, even before a full clinical dataset exists, is standard practice for exactly this reason — it lets the team learn what evidence FDA would want for a Tier-1 claim before designing the Phase 4 prospective study, rather than after.

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8. Workflow and Systems Integration Plan

A model with good statistics that doesn't fit into how a clinic actually works will not reduce time-to-diagnosis. Concretely:

- Where it sits: a patient presents with an adnexal mass on imaging (or persistent symptoms matching a validated symptom index) → today's order set already includes CA-125 ± HE4 → OVAI would run as a reflex or parallel add-on order at that same blood draw, not a new visit or a new specimen type.
- Result delivery: result returned as a structured, discrete value through the existing lab-interface pathway (HL7/FHIR order and result messages) so it appears in the EHR the same way any other send-out lab result does, rather than requiring a separate portal login for the ordering clinician. The standalone Streamlit portal already built is an excellent demo and validation tool, but production delivery should target the clinician's existing EHR, not a side application.
- Decision support: rather than a bare "CRITICAL / LOW RISK" label, the result should map onto existing SGO/ACOG referral guidance for suspected ovarian cancer, ideally as a CDS Hooks-style alert at the point the result lands, so the tool augments a referral decision the clinician is already trained to make rather than introducing a new decision framework.
- Specimen logistics: standard serum tube via the existing phlebotomy workflow, batchable to a central or regional reference lab exactly like any other send-out molecular test — no new collection device required for the near-term Tier-1 claim.
- Hardware track (optional, longer horizon): the prompt explicitly allows hardware solutions. A point-of-care qRT-PCR cartridge for the 5-miRNA signature is a plausible longer-term product but requires materially more capital, its own analytical-validation program, and (likely) a full FDA IVD pathway rather than the LDT route. Recommendation: lead the pitch with the software/reference-lab track, where the data readiness actually is today, and name the point-of-care hardware version as a stated future direction rather than the near-term ask.

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9. How to Present This to the Competition

A concrete package, in priority order:

1. A one-page results scorecard built directly from the Section 4 table — every number labeled with its N and endpoint, the external-validation result shown honestly (ranking AUC good,

threshold transfer currently failing), and one sentence on the funded plan to close that gap (Section 6, Phase 2).

2. A live demo of the existing Streamlit diagnostic portal (single-patient and bulk-upload modes) — this is a real, working, clinician-facing artifact already built, which is rarer at this stage than most competing teams will have. Before demoing it, apply the UI fixes in OVAI_Errors_Suggestions_and_Notes.md §6 (surface the N/endpoint and an "imputed value" flag directly in the interface) so the honesty of the written materials is matched by the honesty of the live tool.
3. A one-page regulatory/reimbursement pathway diagram (Section 7) — this is the single highest-leverage addition relative to what a typical early-stage health-AI pitch shows, because it demonstrates the "genuine understanding of the scope of the problem" the prompt explicitly asks for, beyond model metrics.
4. A one-page workflow diagram (Section 8) showing exactly where in today's clinical pathway the test would be ordered and where its result would surface.
5. An "ask" slide: funding target tied to three concrete, fundable milestones — dbGaP access + HE4 cohort acquisition (Phase 2), a prospective single-platform replication cohort (Phase 2–3), and a CLIA reference-lab partnership to stand up the near-term LDT offering (Section 7).

On Q&A: the team should anticipate "why does your test do worse on the external cohorts?" and have the platform-shift explanation (Section 4, external validation) ready as a rehearsed, confident answer, not a defensive one. In a genuine healthcare-diligence context, a team that has already found, quantified, and explained its own model's biggest weakness — and has a specific, funded plan to fix it — reads as *more* credible than a team presenting only a single high accuracy number, not less.

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10. Metrics Glossary (for non-technical reviewers)

Term	Plain-language meaning
ROC-AUC	How well the model ranks cancer patients above non-cancer patients, from 0.5 (no better than chance) to 1.0 (perfect). Doesn't depend on a specific cutoff.

Sensitivity	Of the women who actually have cancer, what fraction the test correctly flags. The number that matters most for not missing a cancer.
Specificity	Of the women who don't have cancer, what fraction the test correctly clears. The number that matters most for avoiding unnecessary referrals/surgery.
Balanced accuracy	Average of sensitivity and specificity — a fairer single number than plain accuracy when the two groups (cancer/non-cancer) aren't the same size.
Out-of-fold (OOF) / cross-validation	The model is trained on most of the patients and tested only on the ones it never saw during that training round, repeated several times so every patient is tested exactly once.
External validation	Testing the exact frozen model on patients from a completely different study/cohort it never touched during development — the strongest available evidence of real-world generalization.
Calibration	Whether a "70% risk" prediction actually corresponds to about 70% of those patients having cancer, not just whether the ranking is good.
Decision curve / net benefit	Whether using the test to make referral decisions does more good than harm across a realistic range of decision thresholds — the clinical-utility question, as opposed to a pure statistics question.

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11. Immediate To-Do List Before Submission

1. Decide which of the Section 4 numbers is "the" headline number and present the full table regardless.
2. Add the calibration plot and decision-curve analysis (Section 6, Phase 0).

3. Fix the Streamlit portal's transparency gaps (endpoint/N visible next to the benchmark numbers; imputed-value flag on bulk results).
4. Reword every mention of the "gated multimodal neural network" in pitch materials to state plainly that it is untrained architecture, not a working multimodal model.
5. Prepare the one-page regulatory/reimbursement and workflow diagrams (Sections 7–8).
6. Start the dbGaP phs000897 access application now, even though results won't land before the deadline — being able to say "in progress, submitted [date]" is itself evidence of a credible plan.

For the full list of underlying technical findings that this plan is built on — leakage risks, reproducibility gaps, and detailed data/model notes — see the companion document, OVAI_Errors_Suggestions_and_Notes.md.

OVAI — Errors, Suggestions, and Additional Notes

Independent QA pass over the OVAI-main codebase, cross-referenced against the team's own reports/current_pipeline_audit.md (dated 2026-09-19). Findings below extend that audit rather than repeating it — items already fully covered there are only summarized, with new findings given full detail. Severity follows the same scale the team already uses: Critical / High / Moderate / Low.

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1. Reporting and Framing Findings (new — not in the internal audit)

1.1 [High] The headline outline number and the canonical benchmark number are not the same claim

The outline text — *"An engineered logistic regression model... achieved a mean out-of-fold ROC-AUC of 0.8997, with 86.43% balanced accuracy, 82.70% sensitivity, and 90.20% specificity for the evaluated Stage I/II endpoint"* — traces to scripts/test_high_capacity_models.py (→ reports/high_capacity_model_results.md), which evaluates 113 patients: 52 Stage I/II cancers vs. 61 benign/control, with borderline tumors and Stage III/IV cancers excluded. The project's own canonical evaluator (scripts/evaluate_gse94533_diagnostic_model.py) reports 0.813 AUC on the full 179-patient cohort. Both numbers are legitimate, correctly-computed nested-CV estimates — they simply answer different questions, and the gap between them (0.90 vs. 0.81) is mostly explained by which patients are in the denominator, not by a better model. Action: always state N and the exact endpoint next to any percentage in pitch materials; report both numbers side by side (done in the companion methodology document, Section 4).

1.2 [High] The frozen production signature was likely selected using the same 113 patients it is "validated" on

production_artifacts/manifest.json freezes a specific 5-miRNA signature (hsa-miR-200b-3p, hsa-miR-296-3p, hsa-miR-6826-5p, hsa-miR-32-3p, hsa-miR-1255a). That exact set closely matches the miRNAs most frequently selected by the fold-wise L1/elastic-net stability analysis run earlier in the project on the *same* 113-patient early-stage-vs-benign/control endpoint (reports/feature_selection_stability.md, reports/clinical_ablation_results.md). Across the project's own iterations on this fixed cohort, the reported AUC for this endpoint climbs over successive phases:

Phase / script	ROC-AUC (early-stage vs. benign/control)
Phase 6 stability analysis, L1, top-5	0.749
Phase 7 clinical ablation, combined, median imputation	0.827
Phase 8 model/ensemble comparison, regularized logistic	0.823
Phase 9 data augmentation, regularized logistic, no augmentation	0.827
Phase 10 (test_high_capacity_models.py), fixed signature, no interactions ("frozen_logistic")	0.899
Phase 10, fixed signature + 2 interaction terms ("engineered_logistic", shipped model)	0.900

This pattern is consistent with a well-known pitfall in biomarker discovery sometimes called feature-selection leakage or "double-dipping": if the specific 5 miRNAs used in the final frozen model were chosen (even indirectly, across several earlier scripts) using label information from this same 113-patient cohort, then re-running cross-validation on that same cohort with that now-fixed signature will tend to look better than the signature's *true* out-of-sample performance, because the signature was implicitly tuned to this cohort's idiosyncrasies before the "final" evaluation ever ran. This is distinct from — and additional to — the transductive feature-filtering leakage the team's own audit already flags.

This is not a claim of misconduct — it is a normal, easy-to-fall-into consequence of running many exploratory scripts against one small, fixed cohort, and every individual script's internal CV is implemented correctly. But it means the 0.8997 AUC should not be treated as an unbiased estimate of how this exact signature will perform on new patients.

Remediation, either of:

- Re-run feature selection inside every outer fold (as Phases 6–9 already correctly do) using the interaction-term feature space, and report *that* number as canonical instead of the fixed-signature version — this gives an honest nested estimate.

- If the fixed signature is being kept for deployment/interpretability reasons, validate it on a cohort that played no role in choosing it. No such cohort currently exists — this is the same conclusion as the external-validation gap and reinforces why it is the top scientific priority (see methodology document, Section 6).

1.3 [Moderate] The two interaction terms in the production model add ~0.0003 AUC and trade sensitivity for specificity

From the same reports/high_capacity_model_results.md table: the simpler model without interaction terms ("frozen_logistic": Top-5 signature + Age + CA-125 only) scores ROC-AUC 0.8994, balanced accuracy 86.6%, sensitivity 84.6%, specificity 88.5%, versus the shipped model with two interaction terms at ROC-AUC 0.8997, balanced accuracy 86.4%, sensitivity 82.7%, specificity 90.2%. The interaction terms buy essentially no discrimination and cost ~2 points of sensitivity for ~2 points of specificity. For a cancer-detection use case, a missed cancer (false negative) is a materially worse error than an unnecessary referral (false positive), so the simpler, higher-sensitivity model is arguably the clinically preferable choice, not the one currently in production_artifacts/. Recommend revisiting which of the two gets deployed, and stating the sensitivity/specificity tradeoff explicitly rather than only the AUC when justifying the choice.

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2. Leakage / Validity Findings (confirmed by direct code review, summarized from the internal audit)

These were independently re-verified by reading the relevant scripts rather than only trusting the audit's prose — all confirmed accurate:

- [Moderate, confirmed] evaluate_gse94533_diagnostic_model.py and the methylation evaluator filter out non-informative feature columns (notna().any(), nunique() > 1) computed on the full frame before the outer CV split. This uses unlabeled information from held-out rows. It is a small leak (unsupervised, doesn't touch labels) but is real and easy to fix by moving the filter inside each fold.
- [Critical, confirmed] GSE106817/GSE113486/GSE113740 all show 0% sensitivity at the frozen decision threshold, with high apparent "accuracy" driven entirely by class imbalance (each cohort is 85–90%+ non-cancer). Ranking ability (AUC 0.77–0.83) is real; classification at the frozen threshold is not currently usable. Confirmed directly in reports/bulk_external_test_results.md, reports/gse106817_platform_shift_audit.md, and reports/gse106817_phase5.md.

- [High, confirmed] No committed evidence recovers the original historical eLife 135/44 patient train/test split; search_possible_historical_test_splits.py samples 20,000 candidate 44-patient splits and explicitly documents that matching the published AUC does not prove a candidate is the real split. The historical ">80% independent-test" claim should not appear in competition materials as a validated result.
- [Confirmed] TCGA-OV stage endpoint (Stage I/II vs III/IV) scores *below* its own majority-class baseline (78.9% vs. 92.6%) with balanced accuracy 53% and AUC 0.49 — essentially uninformative. This must never be cited as evidence the project can detect cancer; it's a different, currently-unsuccessful question (staging, not detection) that happens to reuse a related data source.

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3. Software Engineering and Reproducibility (new findings)

3.1 [High] No version control in the delivered project

git log on the delivered codebase returns "not a git repository." There is no commit history, so there is currently no way to show *which exact version of the code* produced any given number in a pitch deck, no diff history, and no way to recover an earlier state if something regresses. Recommend initializing git immediately, committing the current state, and tagging the exact commit used to generate every number that goes into competition materials. This is a five-minute fix with outsized credibility value if a judge or investor asks "can I see how this evolved?"

3.2 [Moderate] No LICENSE file

There is no LICENSE anywhere in the repository. Whether this should be permissive open-source, a restrictive/proprietary license, or something in between is a business decision for the team (tied to whether they want to show code publicly on the investor track), but the current state (no license at all) defaults to "all rights reserved" with no clarity for collaborators or judges who may want to inspect the code.

3.3 [Moderate] No automated tests beyond one manual smoke script

scripts/smoke_test.py is the only test in the repository, and it only checks that the untrained multimodal architecture runs and its masking behaves on random tensors — there is no tests/ directory and no pytest usage anywhere. For a project whose core claims rest on "no leakage, no fabricated multimodal rows," a small suite of automated guardrail tests would materially strengthen

the story: e.g., a test asserting zero patient-ID overlap between GSE94533 and every external cohort, a test asserting every reported metric's script uses fold-contained preprocessing, a test asserting the production manifest's feature list matches the model's actual coef_ dimensionality. These are cheap to write and directly demonstrate the rigor claimed in the reports.

3.4 [Moderate, confirmed and extended] Portability: hardcoded paths

45 of the repository's 100 scripts reference Path.home()/ovarian-ai or similar hardcoded locations rather than the configs/project.yaml paths block that already exists for exactly this purpose. A judge or reviewer trying to actually run the delivered zip will hit this on the first legacy script they try. Recommend routing every script through the existing config paths before the deadline, at minimum for whichever scripts are named in competition materials as reproducing a specific number.

3.5 [Moderate] Frozen model artifacts have no recorded library versions

production_artifacts/manifest.json records the feature list, transform, and hyperparameters, but not the scikit-learn / numpy / joblib versions used to fit and serialize frozen_model.joblib. Joblib-pickled scikit-learn estimators are not guaranteed to load correctly across scikit-learn major-version boundaries — this is a real, silent-failure risk if the artifact is loaded in a different environment than it was produced in (e.g., a judge's machine, or a future teammate's laptop). Recommend adding a pip freeze > requirements-lock.txt alongside every frozen artifact, and recording the exact library versions inside manifest.json itself.

3.6 [Low] Extensive near-duplicate script variants

There are multiple clean / standalone / final / _by_id variants of the same historical-reproduction and frozen-signature-transfer experiments (e.g., four separate test_gse8841_signature_on_gse12172*.py scripts, four reproduce_original_historical/pmml*.py scripts). This is technical debt rather than a correctness problem, but it makes the delivered zip hard to navigate for anyone outside the team. Recommend: pick one canonical script per reported result now, move the rest to scripts/legacy/, and add a short RESULTS_INDEX.md that maps every number appearing in competition materials to (script, exact command, output file, N, endpoint) — a very fast, very high-value document to write given how much is already in reports/.

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4. Data and Biological Notes

4.1 [Moderate] No HE4 anywhere in current data assets

HE4 is a component of both ROMA and Overa, the two most relevant FDA-cleared comparators. Its absence means the project currently cannot run a direct, apples-to-apples comparison against either. This is the single most valuable near-term data-acquisition target — recommend actively sourcing an HE4-linked serum cohort, or prioritizing HE4 collection in any prospective data-collection effort.

4.2 [Notable, biologically grounded] False negatives cluster in clear-cell/endometrioid histology, with lower miR-200 family expression

reports/error_analysis.md shows false negatives across every model variant are disproportionately Endo/Clear Cell histology, and the largest expression contrasts between false negatives and correctly-classified cancers are consistently in hsa-miR-200c-3p, hsa-miR-200a-3p, and hsa-miR-29a-3p — all lower in the missed cases. This is not a data artifact; it tracks with published biology: the miR-200 family is tied to epithelial-mesenchymal-transition signaling that is most active in high-grade serous carcinoma, and clear-cell/mucinous histologies are already independently known in the literature to be less well served by CA-125. Recommendation: report performance stratified by histologic subtype as a standing, first-class metric (not just an appendix analysis), and name "does this generalize across histologic subtype" as its own explicit open question in competition materials rather than letting it be smoothed over by the aggregate cancer-vs-non-cancer number. This is a genuine, well-supported scientific insight worth foregrounding — it will read well to a judge with clinical background.

4.3 [Low] Stage I cannot currently be evaluated separately from Stage II

The current GSE94533 metadata group Stage I and Stage II together; every attempt to isolate Stage I alone reports "insufficient class variation." Since Stage I is the highest-value detection target clinically (the point at which 5-year survival is ~90%+), this is worth flagging as a specific data-acquisition or metadata-refinement goal rather than leaving it implicit.

4.4 [Low] CA-125 missingness

CA-125 is missing in roughly 60 of 179 patients (~34%); Age is missing in 1 of 179. Imputation is handled correctly (fold-contained), but given CA-125 is a primary driver of the "combined" model's lift over miRNA-alone, this missingness rate is worth disclosing prominently rather than only in a data-quality appendix.

4.5 [Confirmed, restated for emphasis] The multimodal neural network has never seen real data

src/models/gated_fusion.py is a real, reasonably well-designed architecture (per-modality encoders, learned gates, explicit missing-modality masking), but it has only ever been run against torch.randn

synthetic tensors in scripts/smoke_test.py. This is legitimate scaffold-testing practice, and the README is careful to call it a "scaffold" — but any bullet point or slide describing it should say "engineering scaffold, unit-tested, not yet trained" explicitly, since a reader skimming a feature list ("gated multimodal neural network...") could easily infer real multimodal performance exists.

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5. Product / UX Notes (Streamlit diagnostic portal, scripts/ovarian_diagnostic_portal.py)

- [Moderate] The benchmark numbers (86.43% / 82.70% / 90.20%) are shown prominently in the UI with no N or endpoint definition next to them — only a small caption elsewhere on the page names the feature set. Recommend surfacing "Trained and validated on N=113 (52 early-stage cancer, 61 benign/control); external validation pending" directly adjacent to the benchmark numbers. A clinician-facing tool in a healthcare competition will be judged partly on whether it is honest about its own evidence base *in the interface itself*, not just in a technical appendix nobody in the room will read.
- [Moderate] The risk output uses a fixed 0.50 probability cutoff with a binary "CRITICAL / LOW RISK" label. The team's own enhancement benchmark found that a tuned (non-0.5) threshold improves balanced accuracy meaningfully. Recommend exposing a "high sensitivity" vs. "high specificity" operating-point toggle, with the sensitivity/specificity tradeoff shown, rather than one hardcoded cut presented as if it were the only option.
- [Low] No calibration or confidence indicator is shown alongside the point probability.
- [Low] The bulk-upload path silently substitutes population-median defaults for missing Age/CA-125 but does not flag, in the output results table, which rows used a default vs. a measured value. Recommend adding an "imputed" flag column so a clinician reviewing a batch of results can tell which predictions rest on population defaults.
- [Low] No audit log of predictions made through the portal — worth adding both for internal QA and because prediction logging is standard practice ahead of any future regulatory documentation.

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6. Governance and Compliance Notes

- [Low] No explicit statement anywhere in the repository about the IRB/human-subjects status of using public GEO datasets. This is very likely fine as a legal matter (GEO series are

de-identified and public), but a one-line statement to that effect is standard practice and something a healthcare-focused judging panel or investor will typically expect to see rather than assume.

- [Low] No DATA_SOURCES.md documenting the citation/reuse terms of each GEO series and the eLife supplement individually — most allow reuse with citation, but this should be confirmed and recorded per source rather than assumed.
- [Low, see 3.2] No LICENSE file for the codebase itself.

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7. Prioritized Action List (ranked, before the competition deadline)

1. Present the full endpoint comparison table (Section 1.1 here / Section 4 of the methodology document) rather than the single best number.
2. Fix the two Streamlit portal transparency gaps (endpoint/N visible in-UI; imputed-value flag on bulk results) — cheap, high-credibility fixes.
3. Reword every "gated multimodal neural network" mention in pitch text to state plainly it is untrained scaffold code.
4. Initialize git and commit now, so provenance can be shown for every reported number.
5. Write the one-page RESULTS_INDEX.md mapping headline claims to exact scripts/commands/outputs.
6. Re-run the 113-patient endpoint with feature selection inside the fold (Section 1.2) and have that number ready even if the current one is what's ultimately presented.
7. Add the calibration plot and decision-curve analysis (see methodology document, Section 6, Phase 0).
8. Start the dbGaP phs000897 access application (methodology document, Section 6, Phase 2) — timeline reasons alone justify starting now regardless of the deadline.
9. Decide, and be ready to explain, which of the interaction-term vs. no-interaction-term production models is actually being shipped, given the sensitivity/specificity tradeoff in Section 1.3.
10. Add a LICENSE file appropriate to whichever business/open-source posture the team intends for the investor track.

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8. Longer-Horizon Suggestions (not urgent for the pitch, good research directions)

- Benchmark against other published circulating-miRNA ovarian cancer panels in the literature, not only the project's own historical signature, to contextualize where this signature sits relative to the field.
- Add probability calibration (Platt scaling or isotonic regression) on top of the existing logistic pipeline before any clinical framing of the output probability as a literal risk percentage.
- Treat the platform-shift finding (Section 2) as the project's central scientific result, not a side note: across three independent external cohorts, ranking ability survives (AUC 0.77–0.83) but absolute-scale classification does not — this is a genuinely useful, generalizable finding about circulating-miRNA biomarkers across sequencing/array platforms, and is arguably more publishable and more interesting to sophisticated investors than the internal accuracy number itself.
- Given how consistently every purely statistical fix attempted so far (rank transforms, pairwise ratios, quantile normalization) has failed to rescue external performance, the most credible path forward is prospective, harmonized, multi-site data collection rather than further post-hoc correction of already-collected, heterogeneous public data. This should be the team's central "ask" for both funding and expert clinical collaborators.